Day 32 - [8.3.2] Polynomial Division

Friday, April 21, 2023 12:35 PM

Objective:

- Students will investigate properties of Polynomial equations
  - Polynomial Division

Mathematical Practice(s): reason abstractly and quantitatively, attend to precision, look for and make use of structure

Classwork:

- Polydoku Puzzle
- Desmos (Partner Activity) https://teacher.desmos.com/activitybuilder/custom/67f415e895da22cf16b174f4?utm_campaign=share&utm_content=activity

Homework:

- Day 32 HW

Announcement:

- Day 33: Polynomial Team Task
  - ODD - Monday, 4/28
  - EVEN - Tuesday, 4/29
- Day 34: KC #1 - Polynomials
  - ODD - Wednesday, 4/30
  - EVEN - Thursday, 5/1

BELLringer

Instructions: Solve this polydoku. Write your result in standard form.

	1	2	3	4
A	x		$-4x$	
B	x	x^3		
C	-3			

$+16x$

Standard Form (Sum):

Factored Form (Product):

What do you now know about the roots of this polynomial from your work above?

METHODS AND MEANINGS

MATH NOTES

Polynomial Theorems

Factor Theorem: If $x = a$ is a zero of the polynomial $p(x)$, then $(x - a)$ is a factor of the polynomial, and if $(x - a)$ is a factor of the polynomial, then $x = a$ is a zero of the polynomial.

Example: $x = -3$, $x = 2$, and $x = 1 \pm i\sqrt{7}$ are zeros of the polynomial $p(x) = x^4 - x^3 + 20x - 48$. According to the Factor Theorem, $(x + 3)$,$(x - 2)$, $(x - (1 + i\sqrt{7}))$, and $(x - (1 - i\sqrt{7}))$ are factors of the polynomial.

Fundamental Theorem of Algebra: The Fundamental Theorem of Algebra states that every one-variable polynomial has at least one complex root (remember that every real number is also a complex number). This theorem can be used to show that every polynomial of degree n has n complex roots. This also means that the polynomial has n linear factors, since for every root a, $(x - a)$ is a linear factor.

Note that the total of n roots might include roots that occur multiple times. For example, the third degree polynomial $p(x) = x^3 - 3x^2 + 4$ has three roots—a root at $x = -1$ and a double root at $x = 2$. The three linear factors are $(x + 1)$, $(x - 2)$, and $(x - 2)$.

Integral Zero Theorem: For any polynomial with integral coefficients, if an integer is a zero of the polynomial, it must be a factor of the constant term.

Example: Suppose the integers a, b, c, and d are zeros of a polynomial. Then, according to the Factor Theorem, $(x - a)(x - b)(x - c)(x - d)$ are factors of the polynomial. When you multiply these factors together, the constant term will be $abcd$, so a, b, c, and d are factors of the constant term.

Remainder Theorem: For any number c, when a polynomial $p(x)$ is divided by $(x - c)$, the remainder is $p(c)$. For example, if the polynomial $p(x) = x^4 - x^3 + 20x - 48$ is divided by $(x - 5)$, the remainder is $p(5) = 552$.

Note that it follows from the Remainder Theorem that if $p(c) = 0$, then $(x - c)$ is a factor. For example, if $p(x) = x^3 - 3x^2 + 4$, one solution to $x^3 - 3x^2 + 4 = 0$ is $x = -1$. Since $p(-1) = 0$, then $(x + 1)$ is a factor.